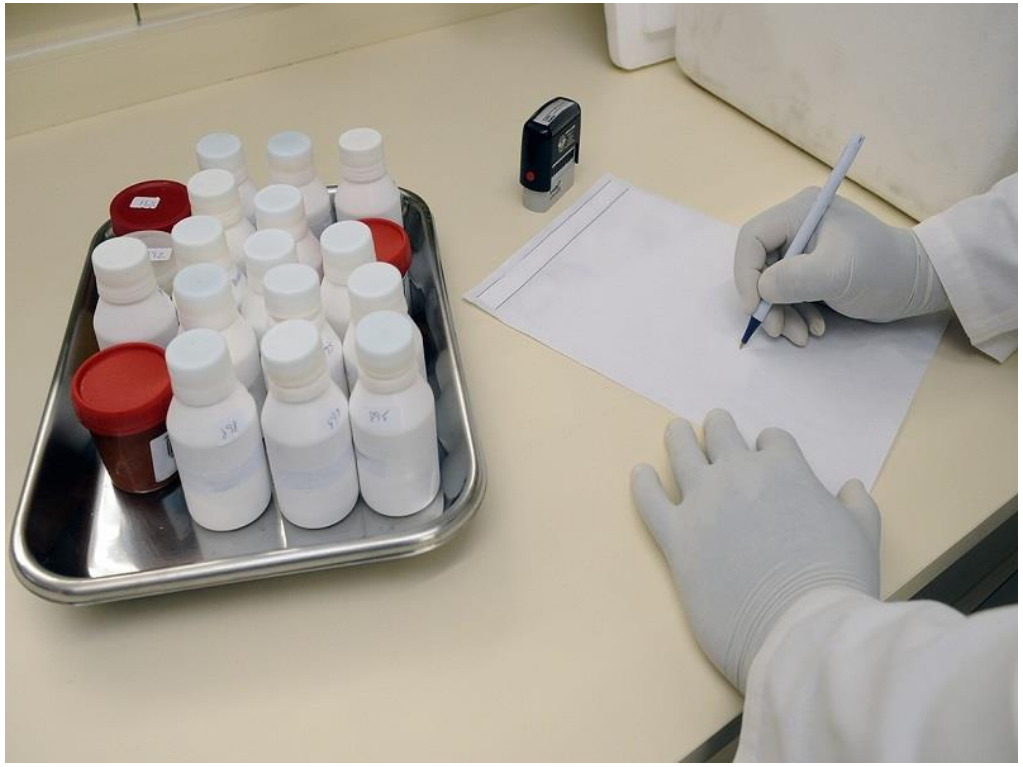
Hypothesis tests for two means or more means



Overview

Taking stock of where we are and where we are going

Hypothesis for two means continued

Hypothesis tests for more than two means

Announcement

Homework 7 has been posted!

It is due on Gradescope on Sunday March 29th at 11pm

The make-up exam will be held during class time this Thursday

- **I.e., March 26th from 2:30-3:45pm in Kline Tower room 1105B**
- **This is the only time the make-up exam will be offered**

We should have the midterm exams graded by Sunday

Any questions before we get started…?

WHERE
ARE WE?

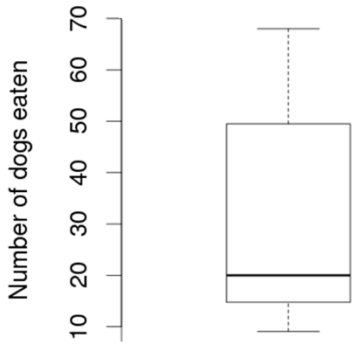
Where Are We
Going?

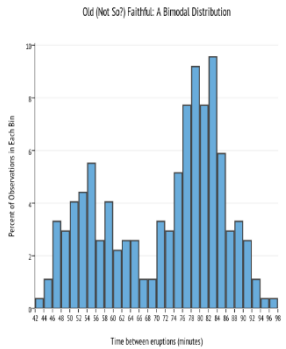
Where we are: what we have covered

Descriptive statistics

Statistical inference

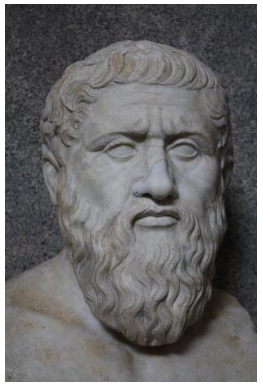
- We have used computational methods for inference

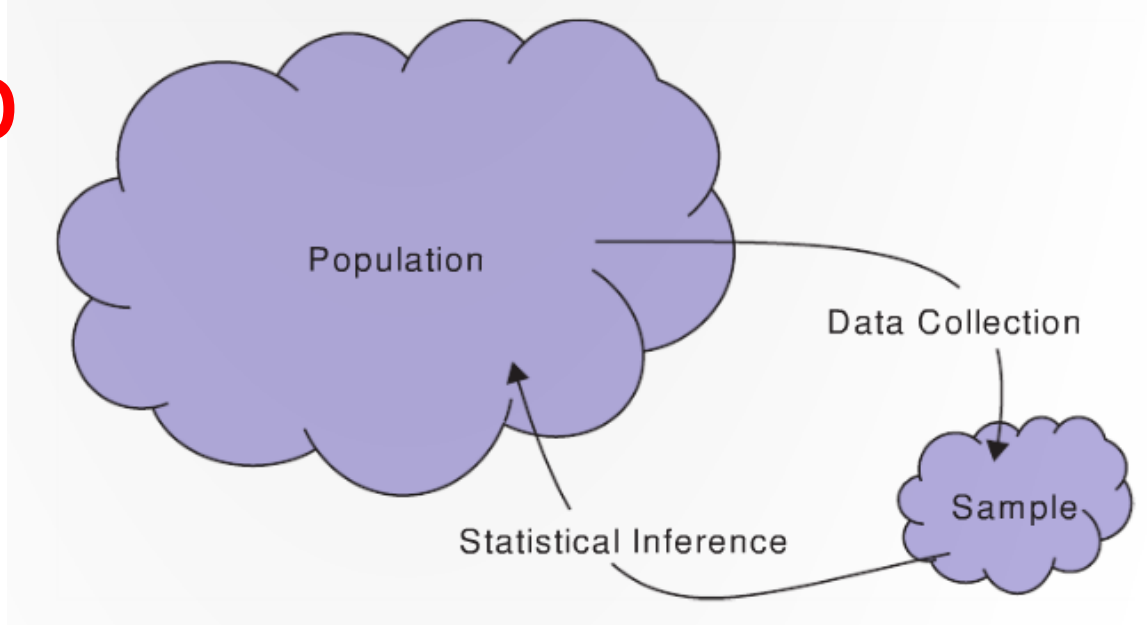




π, μ, σ, ρ

$\hat{p}$, $\bar{x}$, s, r

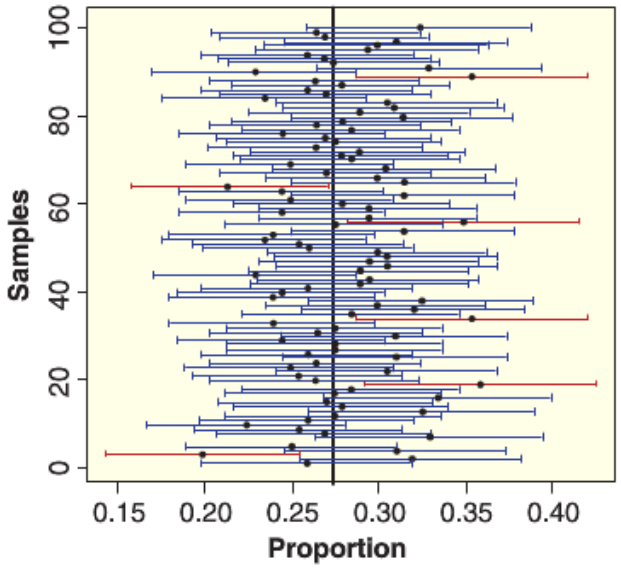




Where we are: what we have covered

Statistical inference using computational methods

- Confidence intervals using the bootstrap
- Hypothesis tests using permutation/randomization tests
  - Single proportion and two means



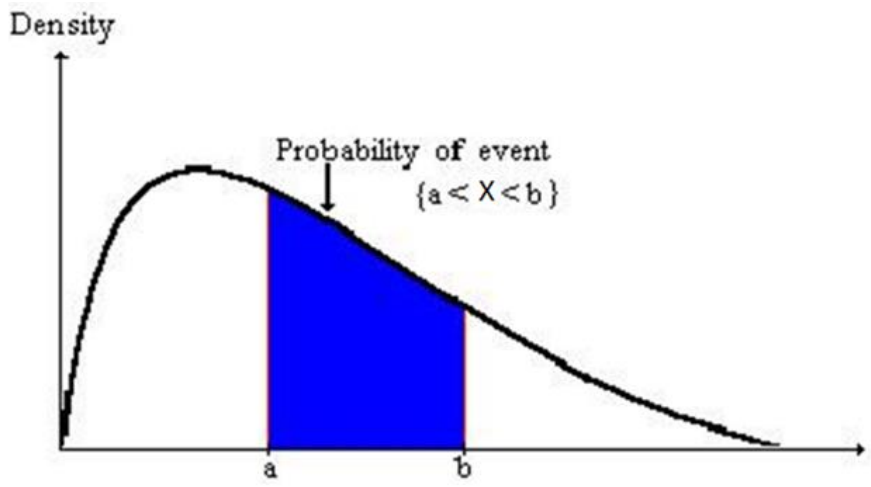
Where we are: where are we going…

Continuation of statistical inference using computational methods

- Hypothesis tests for two or more means and correlation
- Theories of hypothesis tests

Statistical inference based on math/theory

- t-tests, ANOVA, χ^2-test , etc.



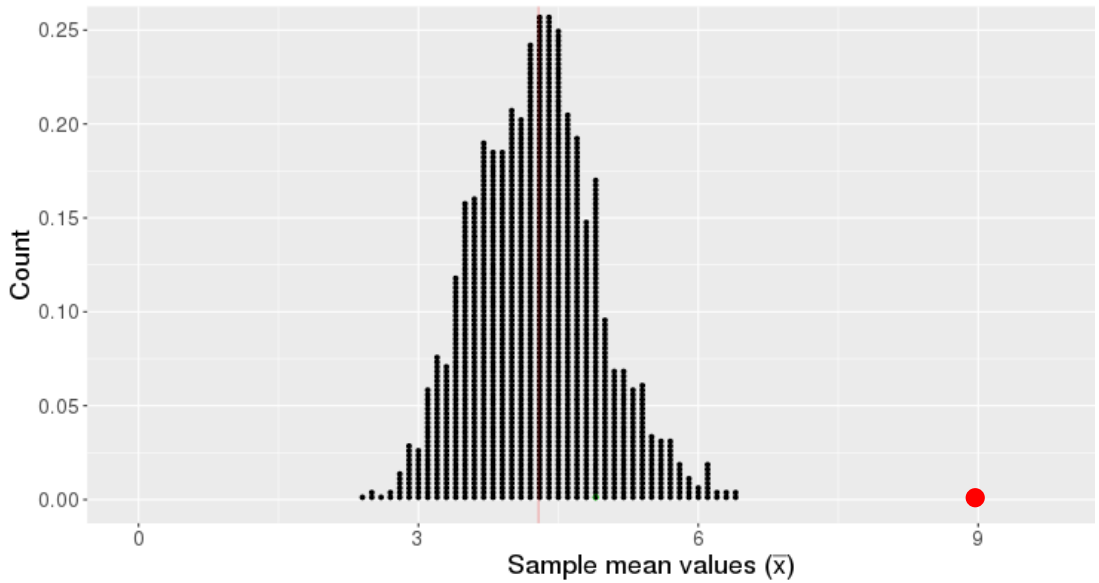
Hypothesis tests for comparing two means

The logic of hypothesis tests…

We start with a claim about a population parameter

- E.g., μ = 4

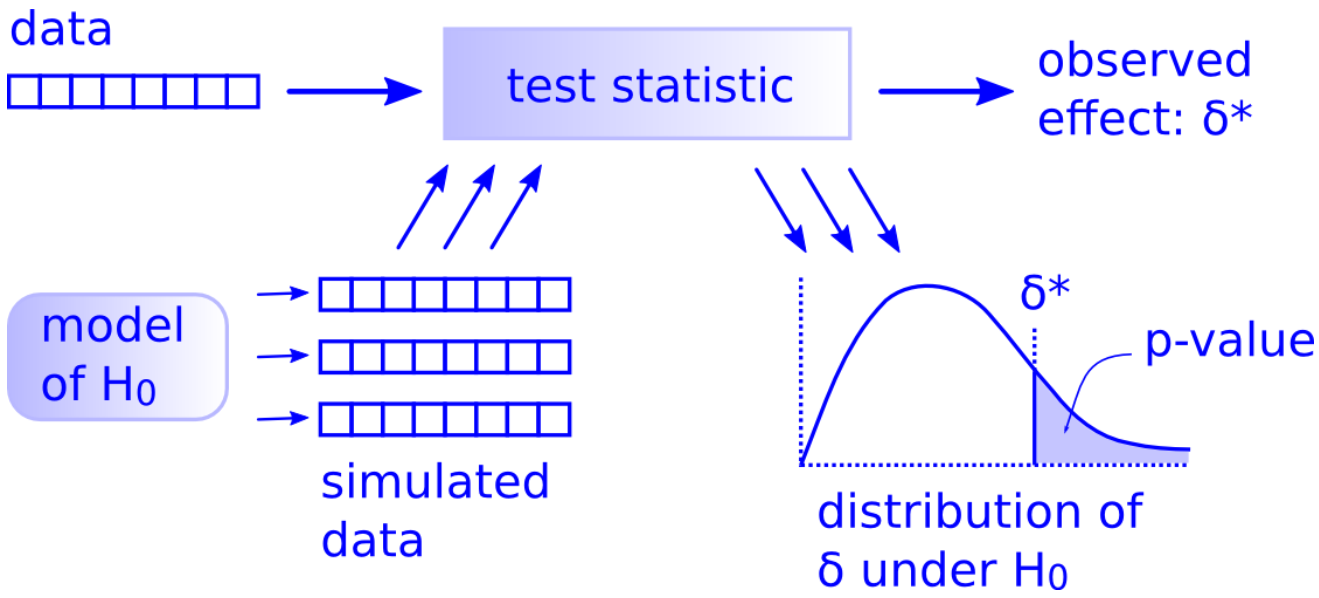
This claim implies we should get a certain distribution of statistics



If our observed statistic is highly unlikely, we reject the claim

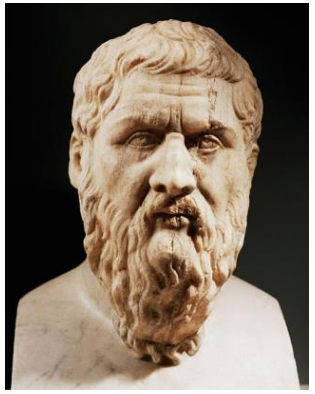
The logic of hypothesis tests…

There is only one hypothesis test!



Just follow the 5 hypothesis tests steps!

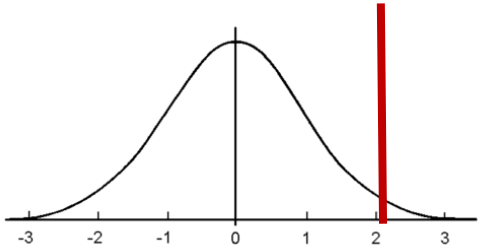
Five steps of hypothesis testing



1. State H_0 and H_A
   - Assume Gorgias (H_0) was right

2. Calculate the actual observed statistic

3. Create a **null distribution** of statistics that are consistent with H_0
   - i.e., a distribution of statistics that we would expect if Gorgias is right



4. Get the probability we would get a statistic more than the observed statistic from the null distribution
   - p-value

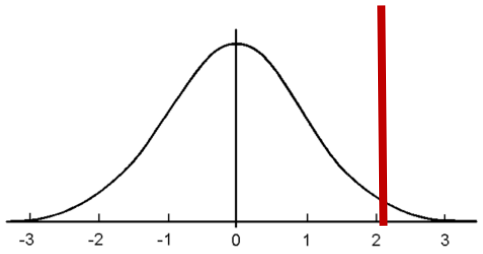
5. Make a judgement
   - Assess whether the results are statistically significant

Five steps of hypothesis testing for proportions

1. State the null and the alternative hypotheses
   - H_0: $\pi = 0.5$ (or another value such as 0.6, etc.)
   - H_A: $\pi > 0.5$

2. Calculate the observed statistic
   - $\hat{p} = .973$

3. Create a null distribution that is consistent with the null hypothesis
   - Null distribution is created by flipping coins: rflip(16, .5)/16



4. Examine how likely the observed statistic is to come from the null distribution
   - pval <- pnull(obs_stat, null_dist, lower.tail = FALSE)

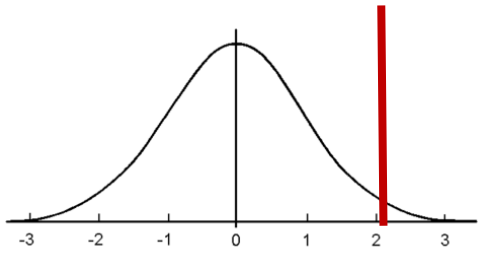
5. Make a judgement
   - If the p-value is small (often less than 0.05) then we can reject the null hypothesis in favor of the alternative hypothesis

Five steps of hypothesis testing for means

1. State the null and the alternative hypotheses
   - H_0: $\mu_T - \mu_C = 0$
   - H_A: $\mu_T - \mu_C > 0$

2. Calculate the observed statistic
   - $\bar{x}_T - \bar{x}_C$

3. Create a null distribution that is consistent with the null hypothesis
   - Null distribution is created by shuffling (to be explained more soon)



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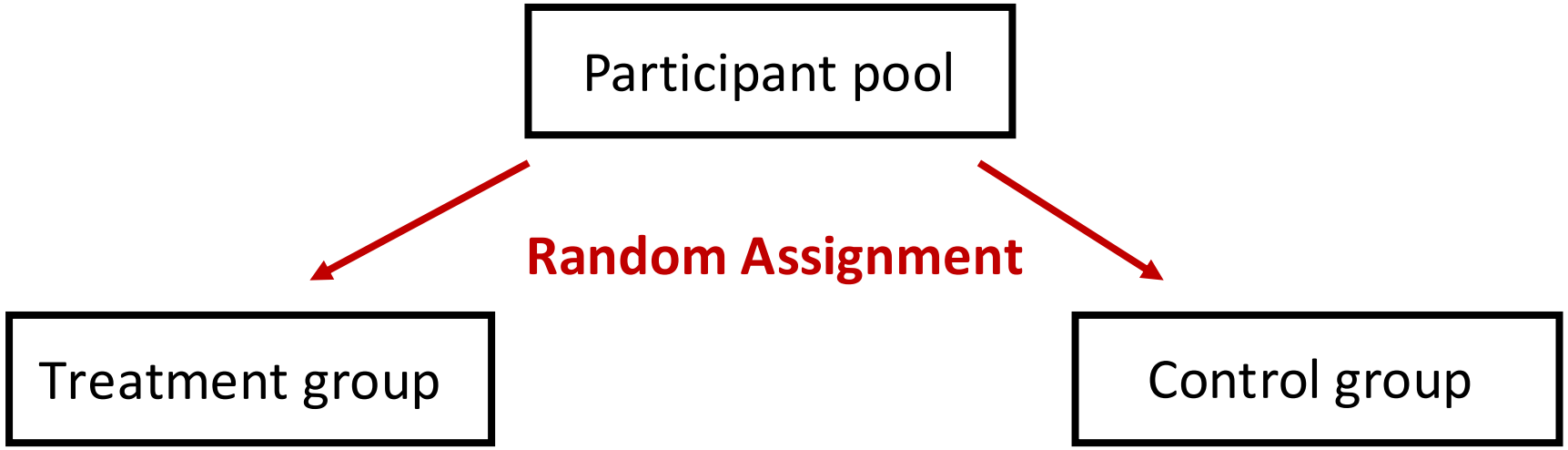
Hypothesis tests for comparing two means

Experimental design: randomized controlled trial

Take a group of participant and ***randomly assign***:

- Half to a *treatment group* where they get the pill
- Half in a *control group* where they get a fake pill (placebo)
- See if there is more improvement in the treatment group compared to the control group



Example: Does calcium reduce blood pressure?

A randomized controlled trial by Lyle et al (1987) investigated whether calcium lowered blood pressure

- A treatment group of 10 men received a calcium supplement for 12 weeks
- A control group of 11 men received a placebo during the same period

The blood pressure of these men was taken before and after the 12 weeks of the study

1. What are the null and alternative hypotheses?

- H_0: $\mu_{Treatment} = \mu_{Control}$ or $\mu_{Treatment} - \mu_{Control} = 0$
- H_A: $\mu_{Treatment} > \mu_{Control}$ or $\mu_{Treatment} - \mu_{Control} > 0$
  - i.e., a <u>greater decrease</u> in blood pressure after taking calcium

Hypothesis tests for differences in two group means

1. State the null and alternative hypothesis

H_0: $\mu_{Treatment} = \mu_{Control}$ or $\mu_{Treatment} - \mu_{Control} = 0$

H_A: $\mu_{Treatment} > \mu_{Control}$ or $\mu_{Treatment} - \mu_{Control} > 0$

2. Write down the statistic of interest using appropriate symbols

$\bar{x}_{Effect} = \bar{x}_{Treatment} - \bar{x}_{Control}$

Does calcium reduce blood pressure?

Treatment data (n = 10):

Begin	107	110	123	129	112	111	107	112	136	102
End	100	114	105	112	115	116	106	102	125	104
Decrease	**7**	**-4**	**18**	**17**	**-3**	**-5**	**1**	**10**	**11**	**-2**

Control data (n = 11):

Begin	123	109	112	102	98	114	119	112	110	117	130
End	124	97	113	105	95	119	114	114	121	118	133
Decrease	**-1**	**12**	**-1**	**-3**	**3**	**-5**	**5**	**2**	**-11**	**-1**	**-3**

2. What is the observed statistic of interest?

- $\bar{x}_{Effect}$ = 5 - -.2727 = 5.273

3. What is step 3?

3. Create the null distribution!

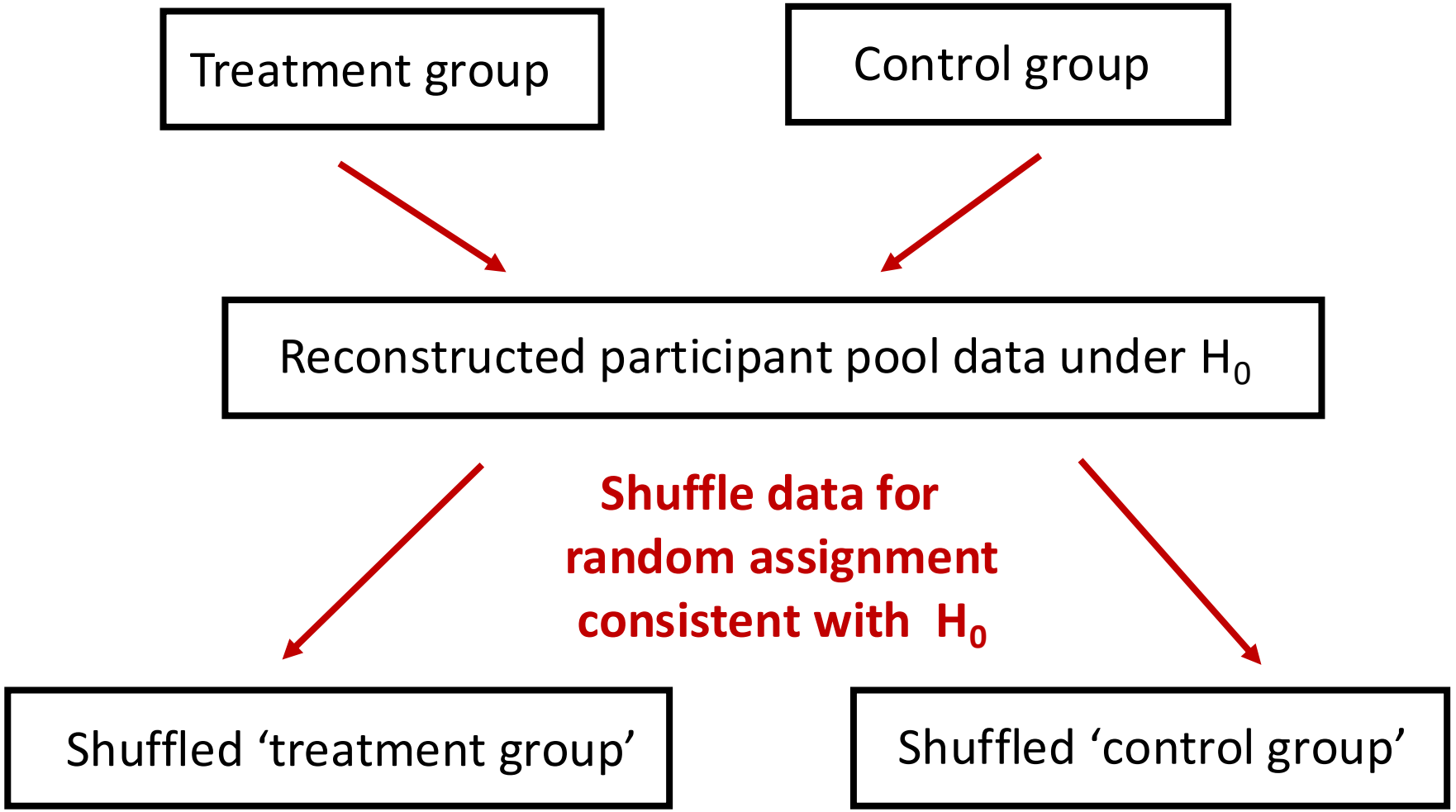
How could we create the null distribution?

Need to generate data consistent with H_0: $\mu_{Treatment} - \mu_{Control} = 0$

- i.e., we need fake $\bar{x}_{Effect}$ that are consistent with H_0

Any ideas how we could do this?

3. Create the null distribution!



One null distribution statistic: $\bar{x}_{Shuff_Treatment} - \bar{x}_{Shuff_control}$

3. Create a null distribution

1. Combine data from both groups

2. Shuffle data

3. Randomly select 10 points to be the shuffled ‘treatment’ group

4. Take the remaining points to the shuffled ‘control’ group

5. Compute the statistic of interest on these shuffled groups

6. Repeat 10,000 times to get a null distribution

3. Creating a null distribution in R

```
# the data from the calcium study
treat <- c(7, -4, 18, 17, -3, -5,  1, 10, 11, -2)
control <- c(-1,  12,  -1,  -3,   3,  -5,   5,   2, -11,  -1,  -3)

# observed statistic
obs_stat <- mean(treat) - mean(control)

#  Combine data from both groups
combined_data <- c(treat, control)
```

3. Creating a null distribution in R

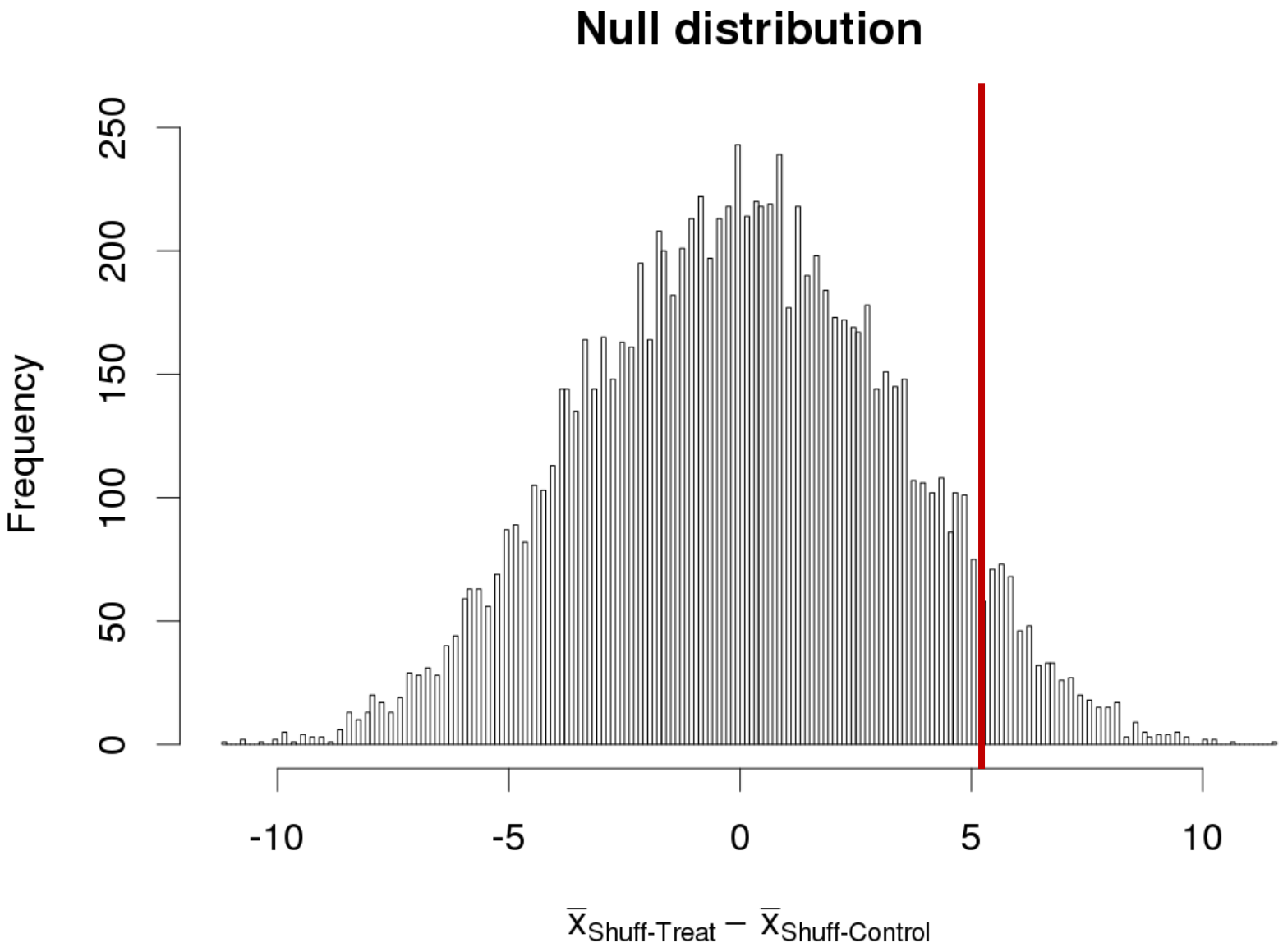
```
null_distribution <-  do_it(10000) * {

    # shuffle data
     shuff_data <- shuffle(combined_data)

     # create fake treatment and control groups
     shuff_treat   <-  shuff_data[1:10]
     shuff_control  <-  shuff_data[11:21]

     # save the statistic of interest
     mean(shuff_treat) - mean(shuff_control)

}
```



hist(null_distribution, breaks = 200)

Next step?

4. Calculate the p-value

```
# Calculate the p-value
    p_value <- pnull(obs_stat, null_distribution, lower.tail = FALSE)
```

p-value = .064

Next step?

5. Are the results statistically significant?

Let’s try it in R…

More practice of hypothesis test for two means

Question: Are husbands older than wives?

To address this question, we can use data from a sample of 100 marriage licenses from St. Lawrence County, NY that give the ages of husbands and wives

Please run the 5 steps of hypothesis testing to assess if the mean age of husbands is greater than the mean age of wives

Let’s try it in R!

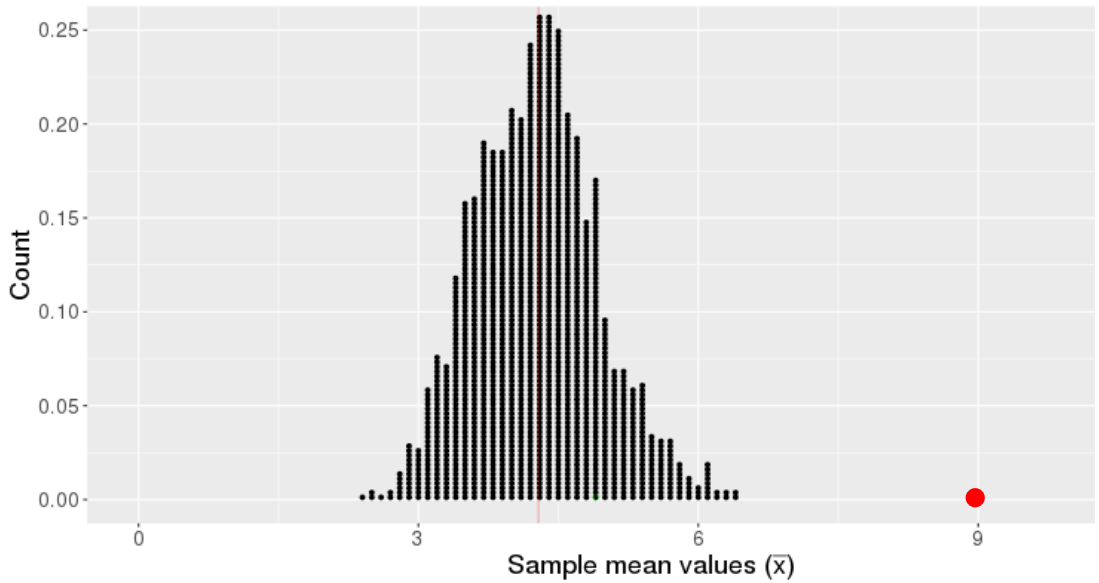
Hypothesis tests for more than two means continued

The logic of hypothesis tests…

We start with a claim about a population parameter

- E.g., μ = 4

This claim implies we should get a certain distribution of statistics



If our observed statistic is highly unlikely, we reject the claim

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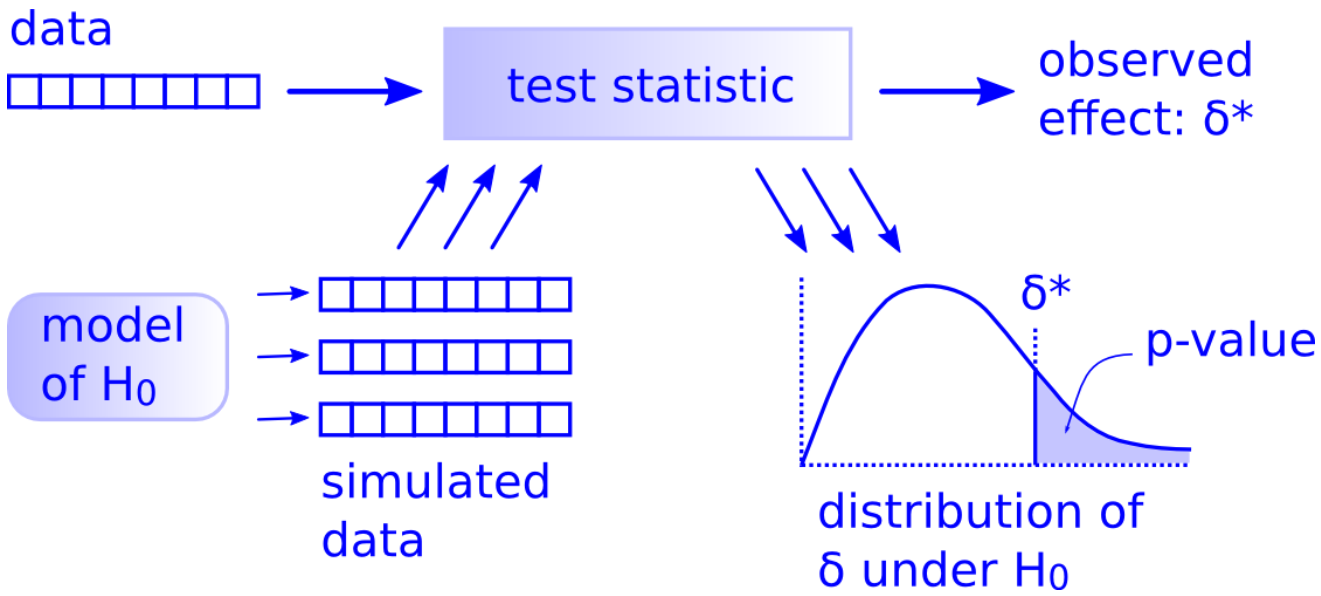
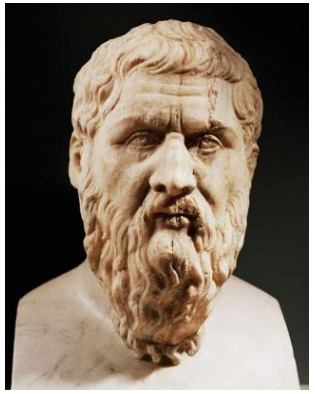


Just follow the 5 hypothesis tests steps!

Five steps of hypothesis testing



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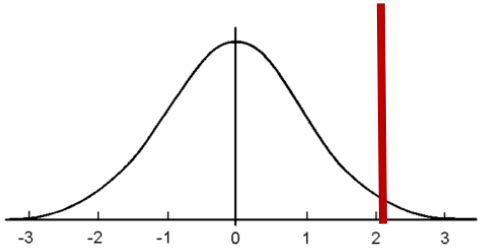

4. Get the probability we would get a statistic more than the observed statistic from the null distribution
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   - Assess whether the results are statistically significant

Comparing more than two means

A group of Hope College students wanted to see if there was an association between a student's major and the time it takes to complete a small Sudoku-like puzzle

	5	3	2		7			8
6		1	5					2
2			9	1	3		5	
7	1	4	6	9	2			
	2						6	
			4	5	1	2	9	7
	6		3	2	5			9
1					6	3		4
8			1		9	6	7	

Comparing more than two means

A group of Hope College students wanted to see if there was an association between a student's major and the time it takes to complete a small Sudoku-like puzzle

They grouped majors into four categories

- Applied science (as)
- Natural science (ns)
- Social science (ss)
- Arts/humanities (ah)

What is the first step of hypothesis testing?

Sudoku by field

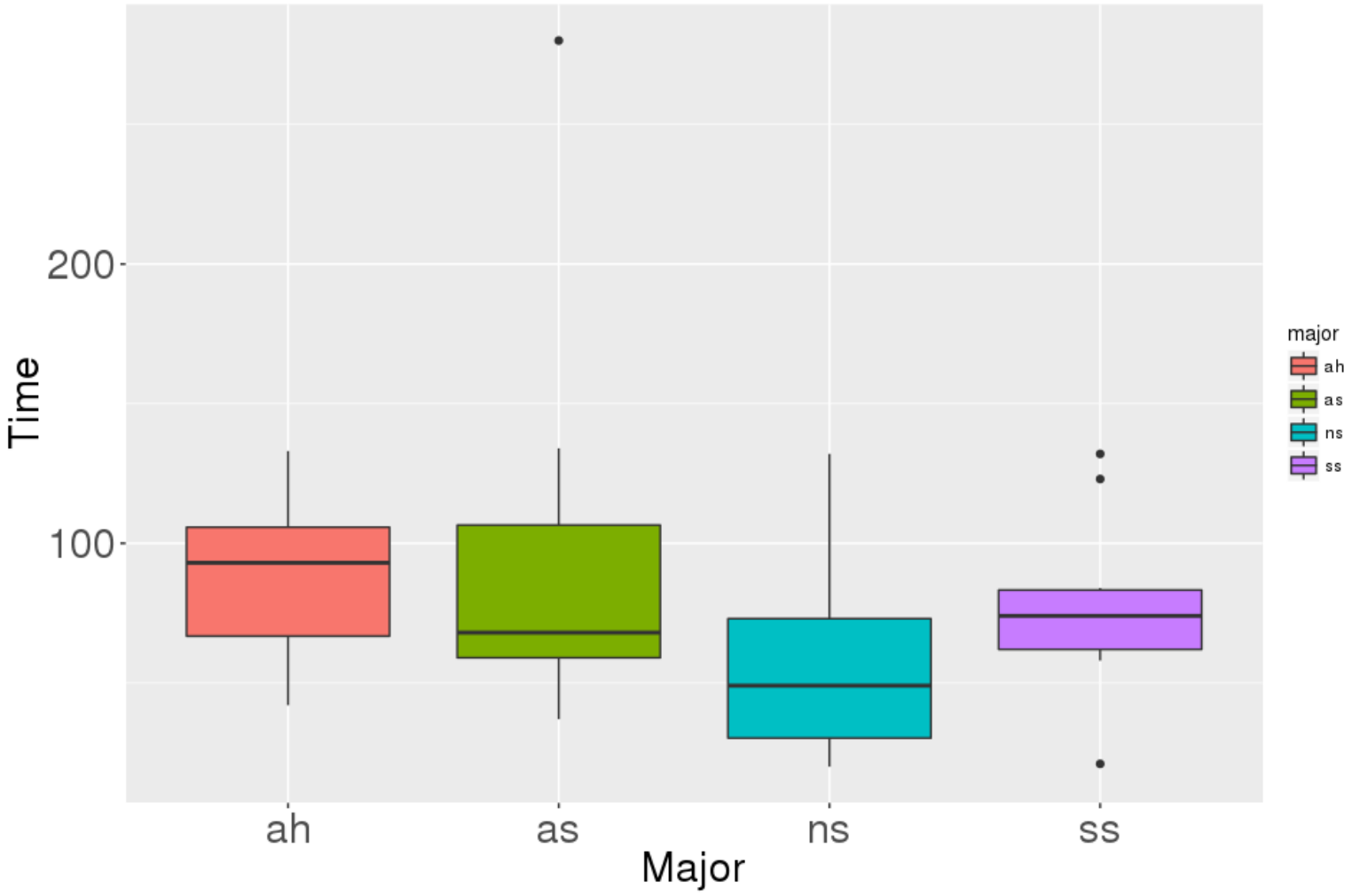
1. State the null and alternative hypotheses!

H_0: $\mu_{as} = \mu_{ns} = \mu_{ss} = \mu_{ah}$
H_A: $\mu_i \neq \mu_j$ for one pair of fields of study

What should we do next?

Let's plot the data first…

Step 2a: Plot of completion time by major



What should we do next?

boxplot(quat_data ~ cat_data)

Sudoku by field

1. State the null and alternative hypotheses!

H_0: $\mu_{as} = \mu_{ns} = \mu_{ss} = \mu_{ah}$
H_A: $\mu_i \neq \mu_j$ for one pair of fields of study

Thoughts on the statistic of interest?

Comparing multiple means

There are many possible statistics we could use. A few choices are:

1. <u>Group range statistic:</u>

   $\max \bar{x} - \min \bar{x}$

2. <u>Mean absolute difference (MAD):</u>

   $(|\bar{x}_{as} - \bar{x}_{ns}| + |\bar{x}_{as} - \bar{x}_{ss}| + |\bar{x}_{as} - \bar{x}_{ah}| + |\bar{x}_{ns} - \bar{x}_{ss}| + |\bar{x}_{ns} - \bar{x}_{ah}| + |\bar{x}_{ss} - \bar{x}_{ah}|)/6$

3. <u>F statistic:</u>

$$F = \frac{\text{between-group variability}}{\text{within-group variability}}$$

Using the MAD statistic

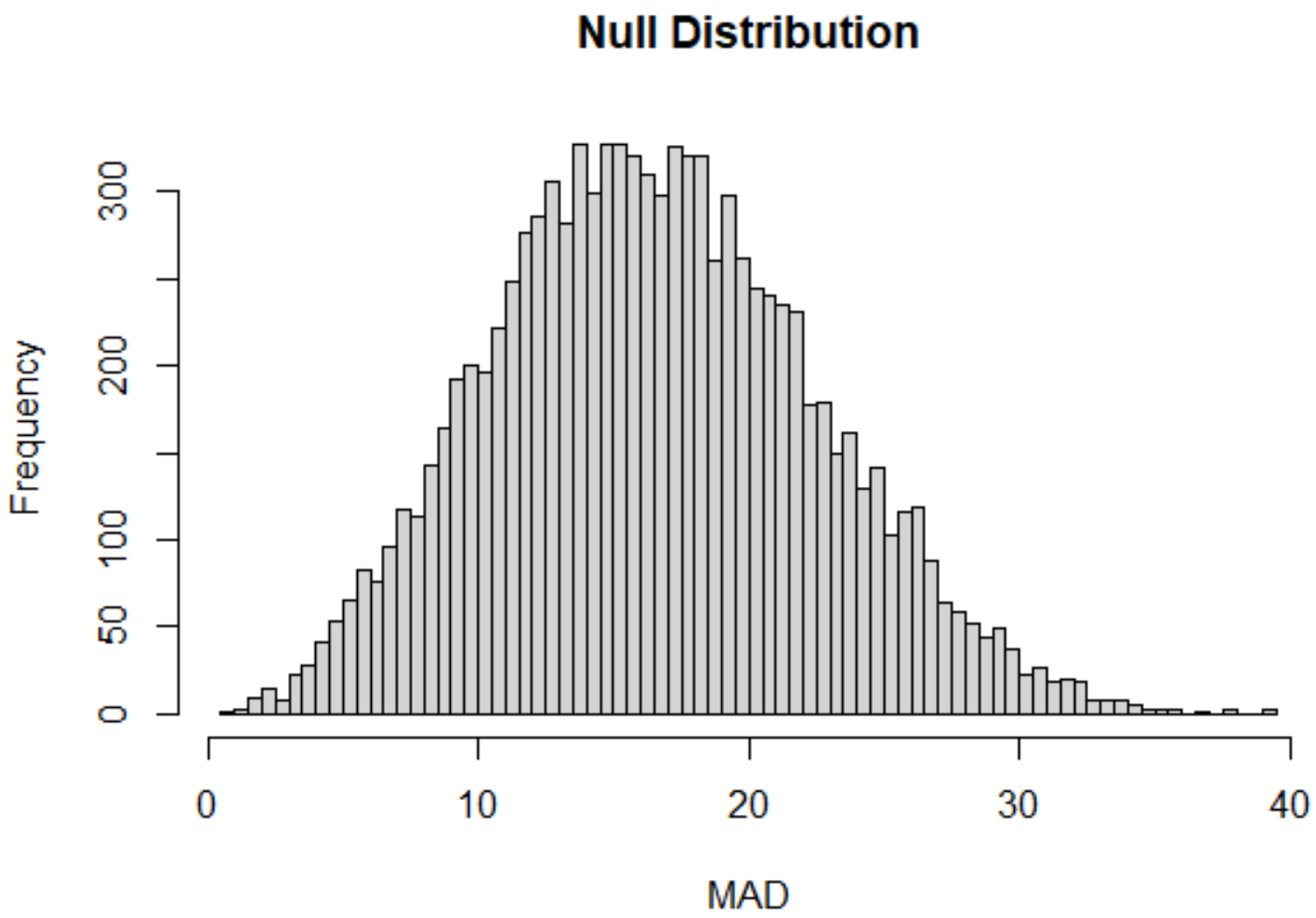
<u>Mean absolute difference (MAD):</u>

$$(|\bar{x}_{as} - \bar{x}_{ns}| + |\bar{x}_{as} - \bar{x}_{ss}| + |\bar{x}_{as} - \bar{x}_{ah}| + |\bar{x}_{ns} - \bar{x}_{ss}| + |\bar{x}_{ns} - \bar{x}_{ah}| + |\bar{x}_{ss} - \bar{x}_{ah}|)/6$$

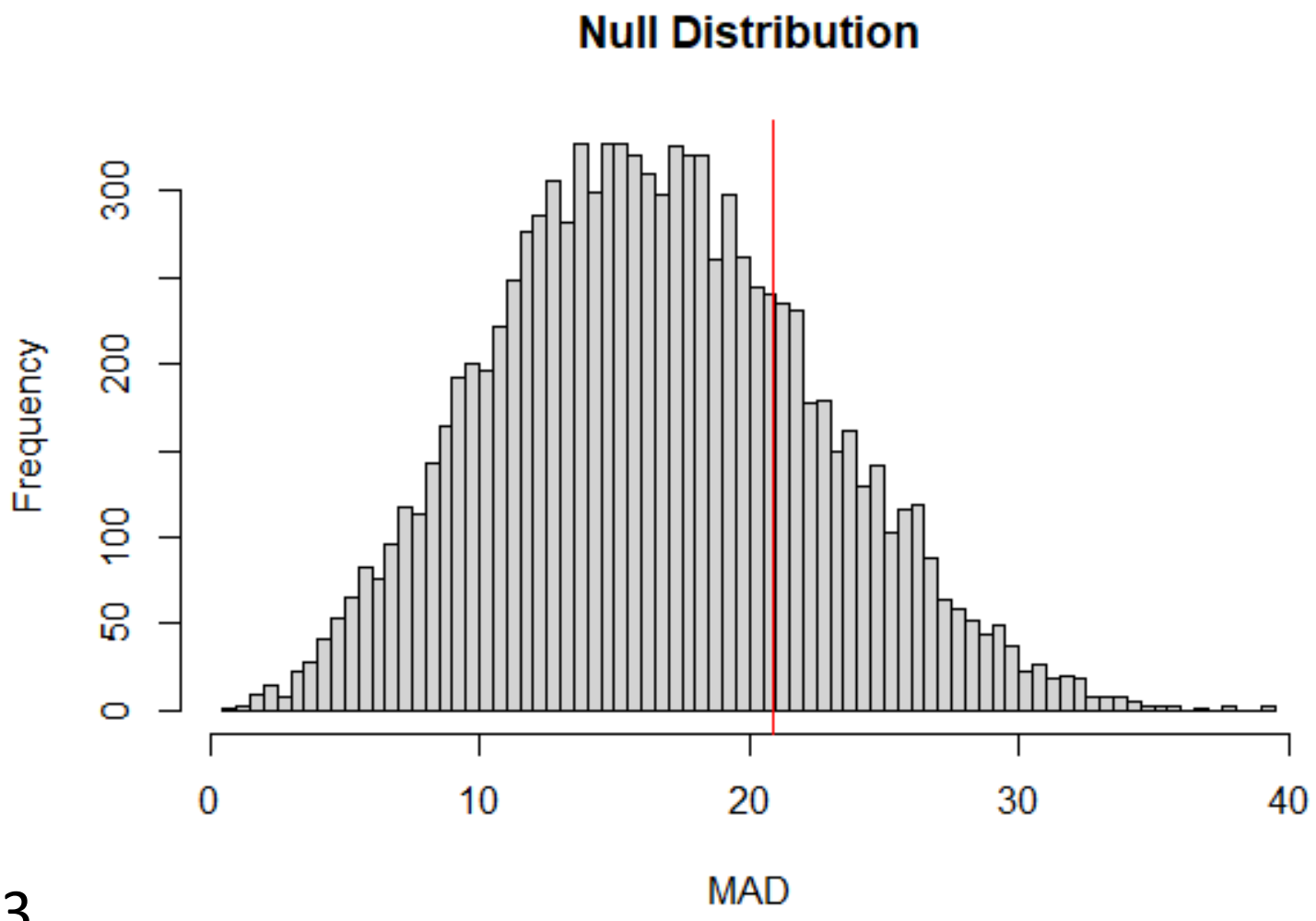
Observed statistic value = 20.88

How can we create the null distribution?

Null distribution



P-value



p-value = .233

Conclusions?

Hypothesis tests for more than two means in R

Hypothesis tests for more than two means in R

Step 1: null and alternative hypotheses…

$\mathbf{H_0}$: $\mu_{as} = \mu_{ns} = \mu_{ss} = \mu_{ah}$

$\mathbf{H_A}$: $\mu_i \neq \mu_j$ for one pair of fields of study

	5	3	2		7			8
6		1	5					2
2			9	1	3		5	
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	2						6	
			4	5	1	2	9	7
	6		3	2	5			9
1					6	3		4
8			1		9	6	7	

Let’s try this analysis in R…

```
# get the data
sudoku_data <- read.table("MajorPuzzle.txt", header = TRUE)

# Extract vectors from the data frame
completion_times <- sudoku_data$time
majors <- sudoku_data$major
```

major	time
ss	21
as	280
ah	96
as	87
ah	133

Visualize the data

How can we visualize the data?

```
# We can create side-by-side boxplots using
boxplot(completion_time ~ major,
        xlab = "Major",   ylab = "Time (s)")
```

major	time
ss	21
as	280
ah	96
as	87
ah	133

Calculating the statistic of interest

We can get the MAD statistic using the get_MAD_stat() function

$$MAD = (|\bar{x}_{as} - \bar{x}_{ns}| + |\bar{x}_{as} - \bar{x}_{ss}| + |\bar{x}_{as} - \bar{x}_{ah}| + |\bar{x}_{ns} - \bar{x}_{ss}| + |\bar{x}_{ns} - \bar{x}_{ah}| + |\bar{x}_{ss} - \bar{x}_{ah}|)/6$$

get_MAD_stat(data_vector, grouping_vector)

- data_vector: a vector of quantitative data
- grouping_vector: a vector of categorical data indicating which group the quantitative data is in

major	time
ss	21
as	280
ah	96
as	87
ah	133

Can you get the MAD statistic for the sudoku data?

```
obs_stat <-  get_MAD_stat(completion_time, major)
```

Creating the null distribution

Q: How could we create one point in a null distribution?

- A: Shuffle the grouping_vector (major vector) and calculate the MAD statistic

Q: How can we do this in R?

```
shuffled_majors <- shuffle(major)

get_MAD_stat(completion_time, shuffled_majors)
```

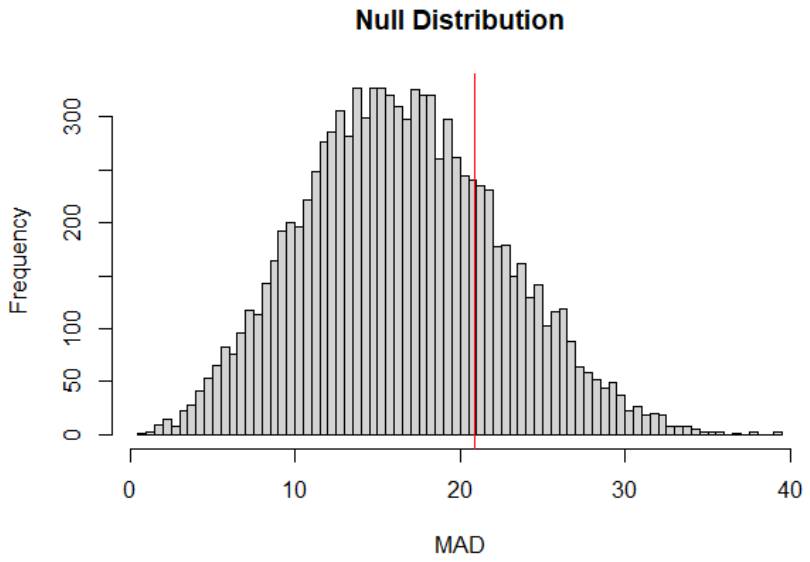
major	time
ss	21
as	280
ah	96
as	87
ah	133

Creating the null distribution

Q: How can we create a full null distribution?

```
null_dist <- do_it(10000) * {
    shuffled_majors <- shuffle(majors)
    get_MAD_stat(completion_times, shuffled_majors)
}

# visualize the null distribution
hist(null_dist, breaks = 200)
abline(v = obs_stat, col = "red")
```

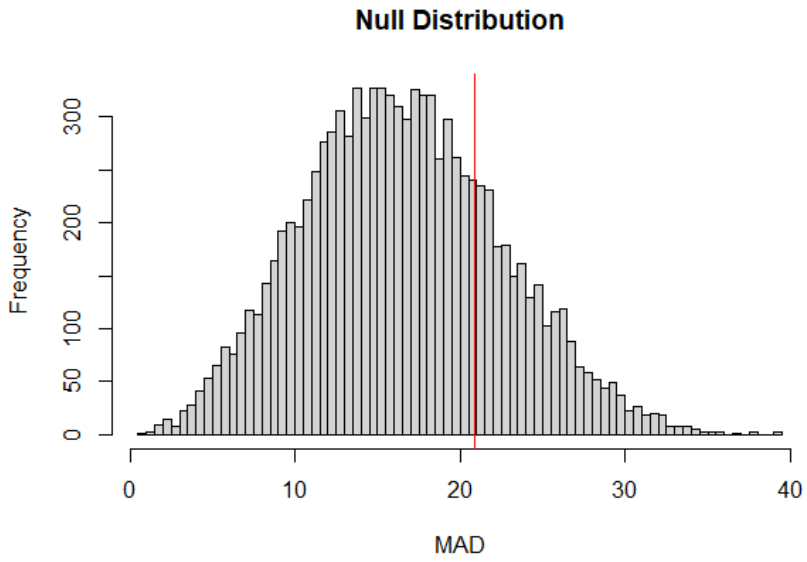


Steps 4 and 5

Q: What do we do next and how do we do it?

- A: We get the p-value

pnull(obs_stat, null_dist, lower.tail = FALSE)

$$MAD = (|\bar{x}_{as} - \bar{x}_{ns}| + |\bar{x}_{as} - \bar{x}_{ss}| + |\bar{x}_{as} - \bar{x}_{ah}| + |\bar{x}_{ns} - \bar{x}_{ss}| + |\bar{x}_{ns} - \bar{x}_{ah}| + |\bar{x}_{ss} - \bar{x}_{ah}|)/6$$

Let's try it in R!